

Model type	Model name	Spearman by Mutational Depth ($\uparrow$)					
		1	2	3	4	5+	All
Alignment-based	Site-Independent	0.336	0.235	0.226	0.267	0.350	0.283
	WaveNet	0.357	0.204	0.250	0.217	0.293	0.264
	EVmutation	0.376	0.274	0.324	0.301	0.394	0.334
	DeepSequence (ens.)	0.405	0.264	0.313	0.309	0.378	0.334
	EVE (ens.)	0.428	0.273	0.308	0.298	0.355	0.332
	GEMME	0.447	0.274	0.321	0.324	0.414	0.356
Protein language	UniRep	0.175	0.071	0.111	0.141	0.191	0.138
	CARP (640M)	0.390	0.213	0.187	0.164	0.162	0.223
	ESM-1b	0.384	0.227	0.187	0.149	0.270	0.243
	ESM-2 (15B)	0.407	0.204	0.239	0.172	0.234	0.251
	RITA XL	0.356	0.139	0.136	0.154	0.233	0.204
	ESM-1v (ens.)	0.403	0.221	0.186	0.151	0.203	0.233
	ProGen2 XL	0.385	0.184	0.280	0.219	0.280	0.270
	VESPA	0.434	0.183	0.357	0.302	0.328	0.321
Hybrid	UniRep evotuned	0.319	0.154	0.250	0.226	0.294	0.249
	MSA Transformer (ens.)	0.426	0.238	0.384	0.366	0.408	0.364
	Tranception L	0.423	0.258	0.352	0.318	0.387	0.348
	TranceptEVE L	0.446	0.280	0.350	0.320	0.382	0.356
Inverse Folding	ESM-IF1	0.439	0.345	0.290	0.289	0.358	0.344
	MIF-ST	0.430	0.265	0.334	0.298	0.298	0.325
	ProteinMPNN	0.292	0.257	0.171	0.186	0.278	0.237

Table A9: **ProteinGym - Zero-shot substitution DMS benchmark by mutational depth** Spearman’s rank correlation between model scores and experimental measurements on the ProteinGym substitution benchmark, separated by mutational depths of 1,2,3,4, and 5 or more. The All column is the average across the 5 depths.

Ranking	Model name	Model type	Spearman
1	ProteinNPT	NPT	0.613
2	Tranception	Embeddings	0.571
3	MSA Transformer	Embeddings	0.568
4	ESM-1v	Embeddings	0.542
5	TranceptEVE	OHE	0.477
6	Tranception	OHE	0.458
7	MSAT	OHE	0.453
8	DeepSequence	OHE	0.440
9	ESM-1v	OHE	0.417
10	OHE w/o augmentation	OHE	0.224

Table A10: **ProteinGym - Supervised substitution DMS benchmark** Ranking based on corrected average of Spearman’s rank correlation between experimental assay measurement and model prediction.

Model type	Model name	Spearman by MSA depth ($\uparrow$)			
		Low	Medium	High	All
NPT	ProteinNPT	0.701	0.587	0.608	0.632
Embeddings	Tranception	0.621	0.556	0.561	0.579
	MSAT	0.685	0.518	0.567	0.590
	ESM-1v	0.653	0.465	0.541	0.553
One-hot encoding	TranceptEVE	0.503	0.483	0.468	0.485
	Tranception	0.490	0.455	0.445	0.463
	MSAT	0.500	0.441	0.448	0.463
	DeepSequence	0.482	0.422	0.426	0.443
	ESM-1v	0.496	0.338	0.400	0.411
	No Augmentation	0.246	0.204	0.227	0.226

Table A11: **Supervised substitution DMS benchmark by MSA depth** Average Spearman’s rank correlation between model scores and experimental measurements by MSA depth on the ProteinGym substitution benchmark. Alignment depth is measured by the ratio of the effective number of sequences N_{eff} in the MSA, following Hopf et al. [2017], by the length covered L (Low: $N_{\text{eff}}/L < 1$; Medium: $1 < N_{\text{eff}}/L < 100$; High: $N_{\text{eff}}/L > 100$)

Model type	Model name	Spearman by Function Type ($\uparrow$)					
		Activity	Binding	Expression	Organismal Fitness	Stability	All
NPT	ProteinNPT	0.577	0.536	0.637	0.545	0.772	0.613
Embeddings	Tranception	0.520	0.529	0.613	0.519	0.674	0.571
	MSAT	0.547	0.470	0.584	0.493	0.749	0.569
	ESM-1v	0.487	0.450	0.587	0.468	0.717	0.542
One-hot encoding	TranceptEVE	0.502	0.444	0.476	0.470	0.493	0.477
	Tranception	0.475	0.416	0.476	0.448	0.473	0.458
	MSAT	0.480	0.393	0.463	0.437	0.491	0.453
	DeepSequence	0.467	0.418	0.424	0.422	0.471	0.440
	ESM-1v	0.421	0.363	0.452	0.383	0.463	0.416
	No Augmentation	0.213	0.212	0.226	0.194	0.273	0.224

Table A12: **Supervised substitution DMS benchmark by function type** Average Spearman’s rank correlation between supervised model scores and experimental measurements on the ProteinGym substitution benchmark, separated into five functional categories. Assays are split into one of Activity, Binding, Organismal Fitness, Stability and Expression. The All column is the average of all the categories

Model type	Model name	Spearman by Taxa ($\uparrow$)				
		Human	Other Eukaryote	Prokaryote	Virus	All
NPT	ProteinNPT	0.649	0.628	0.668	0.580	0.631
Embeddings	Tranception	0.569	0.582	0.594	0.568	0.578
	MSAT	0.634	0.579	0.648	0.521	0.596
	ESM-1v	0.565	0.579	0.617	0.433	0.548
One-hot encoding	TranceptEVE	0.481	0.490	0.475	0.478	0.481
	Tranception	0.457	0.472	0.453	0.456	0.460
	MSAT	0.482	0.459	0.468	0.448	0.464
	DeepSequence	0.451	0.460	0.455	0.383	0.437
	ESM-1v	0.426	0.444	0.452	0.292	0.404
	No Augmentation	0.236	0.217	0.233	0.238	0.231

Table A13: **Supervised substitution DMS benchmark by taxa** Average Spearman’s rank correlation between model scores and experimental measurements on the ProteinGym substitution benchmark, separated into four taxon categories. Assays are split into one of Human, Prokaryote, Other Eukaryote, or Virus. The All column is the average across the categories.

Ranking	Model	Type	Spearman	Std. error
1	ProGen2 M	Protein language model	0.465	0.000
2	ProGen2 Base	Protein language model	0.464	0.010
3	RITA L	Protein language model	0.457	0.034
4	Tranception M no retrieval	Protein language model	0.453	0.036
5*	RITA XL	Protein language model	0.449	0.037
5*	ProGen2 L	Protein language model	0.449	0.011
7	Tranception L no retrieval	Protein language model	0.437	0.041
8	RITA M	Protein language model	0.436	0.030
9	ProGen2 XL	Protein language model	0.431	0.035
10*	TranceptEVE M	Hybrid model	0.424	0.045
10*	ProGen2 S	Protein language model	0.424	0.025
12	TranceptEVE L	Hybrid model	0.412	0.046
13	Tranception S no retrieval	Protein language model	0.410	0.036
14	RITA S	Protein language model	0.397	0.032
15	Tranception L	Hybrid model	0.395	0.043
16	Tranception M	Hybrid model	0.394	0.047
17	HMM	Alignment-based model	0.389	0.045
18	WaveNet	Alignment-based model	0.368	0.067
19	TranceptEVE S	Hybrid model	0.357	0.049
20	PROVEAN	Alignment-based model	0.347	0.046
21	Tranception S	Hybrid model	0.340	0.053
22	ProtGPT2	Protein language model	0.191	0.053
23	UniRep	Protein language model	0.169	0.060

Table A14: **ProteinGym - Zero-shot indel DMS benchmark** Ranking based on corrected average of Spearman’s rank correlation between experimental assay measurement and model prediction. The standard error reported corresponds to the non-parametric bootstrap standard error of the difference between the Spearman performance of a given model and that of the best overall model (ie., TranceptEVE), computed over 10k bootstrap samples from the set of proteins in the ProteinGym DMS indel benchmark.

Dataset	Tranception				TranceptEVE		ProGen2		RITA	WN.	PRO.	HMM
	M	L	M+Ret	L+Ret	M	L	M	XL	L			
DMS Assays												
amyloid β	0.466	0.416	0.442	0.444	0.459	0.466	0.478	0.462	0.438	0.512	0.381	-0.207
β -Lac	0.365	0.344	0.379	0.296	0.401	0.342	0.619	0.409	0.334	0.437	0.385	0.347
AAV	0.371	0.338	0.126	0.210	0.419	0.416	-0.100	0.167	0.103	-0.007	0.177	0.057
Kir2.1	0.437	0.440	0.412	0.391	0.444	0.431	0.432	0.387	0.383	0.408	0.386	0.368
TP53	0.536	0.362	0.579	0.395	0.560	0.399	0.428	0.354	0.383	0.031	0.273	0.482
PTEN	0.678	0.546	0.700	0.563	0.708	0.602	0.552	0.402	0.504	0.697	0.237	0.668
MtrA	0.612	0.395	0.615	0.375	0.562	0.374	0.403	0.348	0.380	0.244	0.278	0.472
OCT1	0.379	0.458	0.447	0.466	0.442	0.453	0.546	0.383	0.542	0.087	0.301	0.280
Tsuboyama	0.169	0.234	0.434	0.463	0.198	0.265	0.511	0.533	0.489	0.475	0.333	0.364
Assays of Natural and Designed Sequences												
AAV	0.677	0.709	0.362	0.691	0.726	0.736	-0.466	0.492	0.543	0.666	0.683	0.607
CM	0.344	0.326	0.219	0.223	0.357	0.340	0.380	0.379	0.337	0.438	0.372	0.398
HIS3	0.611	0.655	0.687	0.707	0.678	0.695	0.702	0.713	0.677	0.687	0.701	0.548

Table A15: **Spearman’s rank correlation between model scores and individual deep mutational scans of indels.** WN and PRO are shorthands for the WaveNet and PROVEAN models respectively.

Ranking	Model	Type	Spearman
1	ESM-1v	Embeddings	0.752
2	Tranception	Embeddings	0.735
3	MSAT	Embeddings	0.689

Table A16: **ProteinGym - Supervised indel DMS benchmark** Ranking based on corrected average of Spearman’s rank correlation between experimental assay measurement and model prediction.

Ranking	Model	Type	AUC
1	ClinPred	Supervised	0.981
2	MetaRNN	Supervised	0.977
3	BayesDel (addAF)	Supervised	0.972
4	VEST4	Supervised	0.929
5	REVEL	Supervised	0.928
6	BayesDel (noAF)	Supervised	0.925
7	VARITY (R)	Supervised	0.921
8	TranceptEVE	Unsupervised	0.920
9	GEMME	Unsupervised	0.919
10	VARITY (ER)	Supervised	0.918
11	EVE	Unsupervised	0.917
12	gMVP	Supervised	0.914
13	CADD	Supervised	0.905
14	PolyPhen2 (HVAR)	Supervised	0.896
15	DEOGEN2	Supervised	0.894
16	ESM-1b	Unsupervised	0.892
17	PROVEAN	Unsupervised	0.886
18	MPC	Supervised	0.881
19	PolyPhen2 (HDIV)	Supervised	0.879
20	SIFT	Unsupervised	0.878
21	SIFT4G	Unsupervised	0.877
22	MutationAssessor	Unsupervised	0.877
23	MutPred	Unsupervised	0.875
24	PrimateAI	Unsupervised	0.855
25	LIST-S2	Unsupervised	0.842
26	MutationTaster	Supervised	0.816
27	DANN	Supervised	0.812
28	LRT	Unsupervised	0.805
29	FATHMM	Supervised	0.723

Table A17: **ProteinGym - Clinical substitution benchmark** Ranking based on AUROC between model prediction and ClinVar benign/pathogenic annotation.

Model Type	Model Name	AUROC(↑)	AUPRC (↑)
Alignment-based models	HMM	0.679	0.775
	PROVEAN	0.926	0.947
	WaveNet	–	–
Protein language models	UniRep	0.395	0.600
	RITA XL	0.923	0.954
	ProGen2 XL	0.846	0.889
	Tranception L (no retrieval)	0.877	0.938
	Tranception M (no retrieval)	0.858	0.929
	ProtGPT2	0.655	0.779
Hybrid models	Tranception L	0.857	0.920
	Tranception M	0.844	0.909
	TranceptEVE	0.857	0.916

Table A18: **ProteinGym - Clinical indels benchmark** Results for indel-compatible baselines on our ClinVar/gnomAD indel benchmark. AUPRC is area under the precision recall curve, and AUROC is area under the receiver-operating characteristic curve. Bold denotes best method, with the runner-up underlined.